

TOPIC 33/60

What is Byte Pair Encoding?

The Algorithm of Language.

LLMs do not read letters, and they do not read words. They read "Tokens." BPE is the exact mathematical algorithm used to create them.

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THE TOKENIZATION DILEMMA

Why Not Words or Letters?

Before feeding text into a neural network, we must convert it into numbers. Historically, engineers tried two extremes, both of which failed at scale.

Word-Level

Dictionary Size: Infinite

If every word gets a unique ID, what happens with typos ("helo"), slang, or compounding words ("un-fathom-able")?

Result: Massive Out-of-Vocabulary (OOV) errors.

Character-Level

Dictionary Size: ~256 (Small)

If we tokenize by character, the word "Apple" becomes 5 tokens. This wastes the LLM's context window and destroys the inherent meaning of the whole word.

Result: Context limits reached too quickly.

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THE SOLUTION

Subword Tokenization

Byte Pair Encoding (BPE) is a data compression algorithm introduced in 1994. OpenAI adapted it for NLP to find the perfect middle ground between characters and words.

The BPE Philosophy:

- ✓ **Frequent words** remain intact as single tokens.
Example: "Apple" = [Apple] (1 token)
- ✓ **Rare words** are broken down into recognizable subword chunks.
Example: "Neuroplasticity" = [Neuro] [plastic] [ity] (3 tokens)
- ✓ **Unknown words (typos)** fall back to individual characters.
Example: "xzyq" = [x] [z] [y] [q] (4 tokens)

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ALGORITHM: PHASE 1

Start with Characters

To build a BPE vocabulary, we first take a massive training corpus (like all of Wikipedia) and break every single word down into its base bytes (characters).

TRAINING CORPUS EXTRACT

"lowest"

INITIAL STATE (CHARACTERS ONLY)

l o w e s t

Our dictionary currently only knows individual letters. Vocabulary size = 26.

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ALGORITHM: PHASE 2

Find the Most Frequent Pair

The algorithm scans the entire dataset to find the two adjacent tokens that appear next to each other most frequently.

Pair Frequencies	Count across corpus
e + r	1,200 times
i + n	1,800 times
e + s	3,500 times 🏆



ALGORITHM: PHASE 3

The Merge Operation

Once the most frequent pair is found, BPE merges them into a brand new, single token. It adds this new token to its dictionary, and replaces every instance of the pair in the dataset.

e **s** → **es**

UPDATING "LOWEST"

[ɪ] [o] [w] [e] [s] [t] becomes [ɪ] [o] [w] **[es]** [t]

The vocabulary size increases by 1. Sequence length decreases.

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PUTTING IT TOGETHER

Iterative Compression

The algorithm repeats this process tens of thousands of times. Subwords combine into larger subwords, which eventually combine into full, common words.

ITERATION 0

[l] [o] [w] [e] [s] [t]

ITERATION 100

[l] [o] [w] [es] [t]

ITERATION 500

[l] [o] [w] [est]

ITERATION 5000

[low] [est]

ITERATION 15000

[lowest]



EDGE CASES

The "Glitch Token" Anomaly

If a dataset contains bizarre, repeated text (like Reddit usernames scraped millions of times), BPE will merge them into a single token to save space.

KNOWN BUG

Token ID: 42168

"SolidGoldMagikarp"

In early GPT models, a Reddit user named *SolidGoldMagikarp* was mentioned so frequently in the training data that BPE made it a single token. However, the model never learned the *meaning* of this token. If you asked GPT-3 about it, it would panic, hallucinate, or spout nonsense because it had a "word" in its vocabulary with no semantic definition attached.

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ECONOMIC IMPACT

The Multilingual Penalty

Because most LLMs are trained heavily on English text, BPE performs millions of merges for English words, compressing them highly. Non-English languages are left fragmented.

TOKENIZATION COMPARISON

English

I love apples

3

Tokens

Hindi (मुझे सेब पसंद हैं)

म उ झ े स े
ब पस ं द ह ै
ं

13

Tokens

Result: Using an LLM in Hindi costs ~4x more money via API and eats context windows faster.

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BPE in Modern Models

VOCABULARY SCALE



GPT-4 uses the tiktoken (cl100k_base) tokenizer.



Its dictionary size is exactly 100,277 tokens.



BPE balances Context Window size vs. Dictionary Size.



To an LLM, " token", "Token", and "token" are different IDs.

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